

PORTLAND HP, United Kingdom

WMO: 038580

Lat: 50.565N

Lon: 2.449W

Elev: 10

StdP: 14.69

Time Zone: 0.00 (EUW)

Period: 88-99

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	30.9	33.9	23.6	17.8	35.5	26.5	20.4	37.5	40.9	48.2	36.1	47.8	18.7	50	0.685

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.5	73.3	63.2	70.3	62.1	68.2	61.3	64.9	69.8	63.8	67.9	62.9	66.7	10.0	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.3	87.0	66.4	62.2	83.7	65.5	61.2	80.7	64.8	29.9	70.3	29.1	67.9	28.4	66.9	72.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 34.5	31.0	27.8	DB	28.1	77.9	4.5	4.3	24.9	81.0	22.3	83.5	19.7	86.0	16.5	89.1	(4)
(5)			WB	26.4	68.3	4.5	2.5	23.2	70.1	20.6	71.6	18.1	73.0	14.8	74.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.9	45.0	44.5	47.0	48.5	54.6	58.5	62.1	63.6	59.8	55.2	49.7	45.8	(6)	
	DBStd	7.86	5.06	4.90	3.23	3.52	4.63	3.50	3.00	3.26	2.89	4.45	4.96	5.51	(7)	
	HDD50	716	159	156	99	67	9	0	0	0	0	12	66	148	(8)	
	HDD65	4449	620	574	557	495	323	200	100	66	157	302	460	595	(9)	
	CDD50	1779	4	2	6	22	152	254	376	420	294	175	57	17	(10)	
	CDD65	39	0	0	0	0	1	4	11	22	1	0	0	0	(11)	
	CDH74	63	0	0	0	0	1	8	12	41	1	0	0	0	(12)	
	CDH80	3	0	0	0	0	0	0	0	3	0	0	0	0	(13)	
(14) Wind	WSAvg	13.8	17.3	15.5	14.4	12.6	11.5	11.3	11.3	11.2	13.2	16.0	14.6	17.4	(14)	
(15) Precipitation	PrecAvg	32.9	3.8	2.8	2.4	2.1	2.0	2.0	1.8	2.3	2.5	3.7	3.9	3.7	(15)	
	PrecMax	44.4	7.6	6.5	5.1	5.4	4.4	4.8	4.8	5.1	5.1	5.6	8.1	6.8	(16)	
	PrecMin	26.3	0.7	0.4	0.1	0.1	0.3	0.3	0.2	0.4	0.5	1.3	0.9	1.1	(17)	
	PrecStd	4.7	1.7	1.5	1.0	1.3	1.1	1.1	1.1	1.3	1.3	1.3	1.7	1.7	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.8	53.9	56.9	61.8	71.9	74.4	75.9	78.2	70.2	65.7	59.2	55.7	(19)	
		MCWB	52.0	51.2	51.5	53.2	60.2	62.1	64.7	64.7	62.7	60.5	56.5	53.4	(20)	
	2%	DB	52.5	52.2	53.6	57.5	67.9	69.8	72.2	73.9	67.1	63.2	58.2	54.8	(21)	
		MCWB	51.0	50.1	49.4	51.5	58.6	60.0	63.0	64.2	60.8	58.7	55.8	52.8	(22)	
	5%	DB	51.8	51.2	52.2	55.1	64.4	66.4	69.6	70.4	65.6	61.9	57.3	53.9	(23)	
		MCWB	50.3	49.2	48.7	49.9	56.3	58.4	62.0	63.1	60.0	58.1	55.2	52.2	(24)	
	10%	DB	51.0	50.3	51.2	53.3	61.4	63.8	67.4	68.3	64.3	60.9	56.3	53.0	(25)	
		MCWB	49.4	48.2	48.2	49.1	55.1	57.5	61.0	62.2	59.4	57.6	54.2	51.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.7	52.2	52.1	54.9	62.3	63.2	66.4	67.7	64.6	61.3	57.5	54.6	(27)	
		MCDB	53.0	52.9	54.5	59.5	69.2	72.3	72.9	73.0	67.9	63.4	58.4	55.2	(28)	
	2%	WB	51.6	51.1	50.9	52.3	59.4	61.0	64.4	65.4	62.9	60.2	56.6	53.6	(29)	
		MCDB	52.2	51.9	52.7	55.5	66.1	67.3	69.7	70.5	65.1	62.2	57.6	54.3	(30)	
	5%	WB	50.8	49.9	50.0	51.1	57.3	59.6	63.0	64.4	62.0	59.3	55.6	52.7	(31)	
		MCDB	51.5	50.9	51.3	53.6	63.2	64.8	67.5	68.2	64.1	61.2	57.0	53.6	(32)	
	10%	WB	49.9	48.6	49.2	50.2	55.7	58.3	61.9	63.5	61.0	58.3	54.6	51.5	(33)	
		MCDB	50.8	49.9	50.6	52.3	60.4	62.7	65.8	67.3	63.4	60.4	56.1	52.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.7	6.3	6.6	8.3	9.2	8.4	8.0	8.5	7.9	6.8	6.3	6.0	(35)	
		MCDBR	5.3	6.5	7.3	10.9	13.7	13.3	12.1	11.7	9.1	6.9	5.3	5.0	(36)	
	5% WB	MCWBR	6.0	6.3	5.8	7.0	7.5	7.5	7.0	6.7	6.3	6.2	5.5	5.7	(37)	
		MCWBR	5.4	6.1	5.5	8.7	12.3	11.5	9.9	9.3	7.1	6.5	5.4	5.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.335	0.350	0.385	0.400	0.403	0.396	0.390	0.383	0.374	0.363	0.335	0.325	(40)	
		taud	2.375	2.386	2.292	2.280	2.317	2.369	2.389	2.430	2.440	2.438	2.425	2.352	(41)	
	Ebn at Noon		217	247	257	267	271	273	273	270	259	239	216	200	(42)	
		Edh at Noon	21	26	34	38	39	37	36	33	29	25	20	18	(43)	
(44) All-Sky Solar Radiation	RadAvg		259	495	885	1394	1684	1826	1735	1452	1096	612	328	206	(44)	
	RadStd		23	54	112	124	89	180	126	114	70	45	32	25	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (64 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page